Assumptions: Since no order number is provided and no additional fields may be added, the combination of customerNumber, orderDate, and orderTime uniquely identifies an order. customerNumber uniquely identifies a customer. partNumber uniquely identifies a part. One customer can make many orders. One order can only be made by one customer. One order can contain multiple parts. One part can be inside many different orders. Each part has one current cage location and one standard unit price. One employee can help multiple customers in making orders. One customer can be helped by only one employee to make a certain order.

This is the original form of the table with no normalization. The information is repeating because one order form can have many parts. The partNumber, partName, partType, cageCode, quantityOrdered, and unitPrice is repeating group since it's basically saying part1 part2 part3 part4. So it's not even 1NF.

parts_order_form

customerNumber	cutomerName	customerType	employeeName	orderTime	orderDate	partNumber	partName	partType	cageCode	quantityOrdered	unitPrice
HG54587	Jeff Peterson	Consumer	D. Harrison	10:30am	7/1/2024	10654 10456 10776 10657	Float Control Modulator Hose Assembly Float Assembly	Plumbing Electrical Plumbing Plumbing	G413 H433 G413 G413	4 3 7 5	12 7 9 10

A customer cannot place more than one order at exactly the same date and time. Therefore, customerNumber, orderDate, and orderTime together uniquely identify an order. This is 1NF because the repeating group has been taken cout so that each cell is atomic value, but there are still partial dependencies like customerName, customerType, partName, etc. that only rely on customerNumber or partNumber, not the full composite PK.

order_form

customerNumber (PK)	**orderTime (PK)**	**orderDate (PK)**	customerName	customerType	employeeName
HG54587	10:30am	7/1/2024	Jeff Peterson	Consumer	D. Harrison

order_part

customerNumber (PK)	**orderTime (PK)**	**orderDate (PK)**	**partNumber (PK)**	partName	partType	cageCode	unitPrice	quantityOrdered
HG54587	10:30am	7/1/2024	10654	Float Control	Plumbing	G413	12	4
HG54587	10:30am	7/1/2024	10456	Modulator	Electrical	H433	7	3
HG54587	10:30am	7/1/2024	10776	Hose Assembly	Plumbing	G413	9	7
HG54587	10:30am	7/1/2024	10657	Float Assembly	Plumbing	G413	10	5

Now there're no partial dependencies because I separated all the dependent attributes with their rely PK to individual tables. The tables now do not have repeating group or partial dependencies so it's 2NF.

part_info

partNumber (PK)	partName	partType	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

order_part

customerNumber (PK)	**orderTime (PK)**	**orderDate (PK)**	**partNumber (PK)**	quantityOrdered
HG54587	10:30am	7/1/2024	10654	4
HG54587	10:30am	7/1/2024	10456	3
HG54587	10:30am	7/1/2024	10776	7
HG54587	10:30am	7/1/2024	10657	5

customer

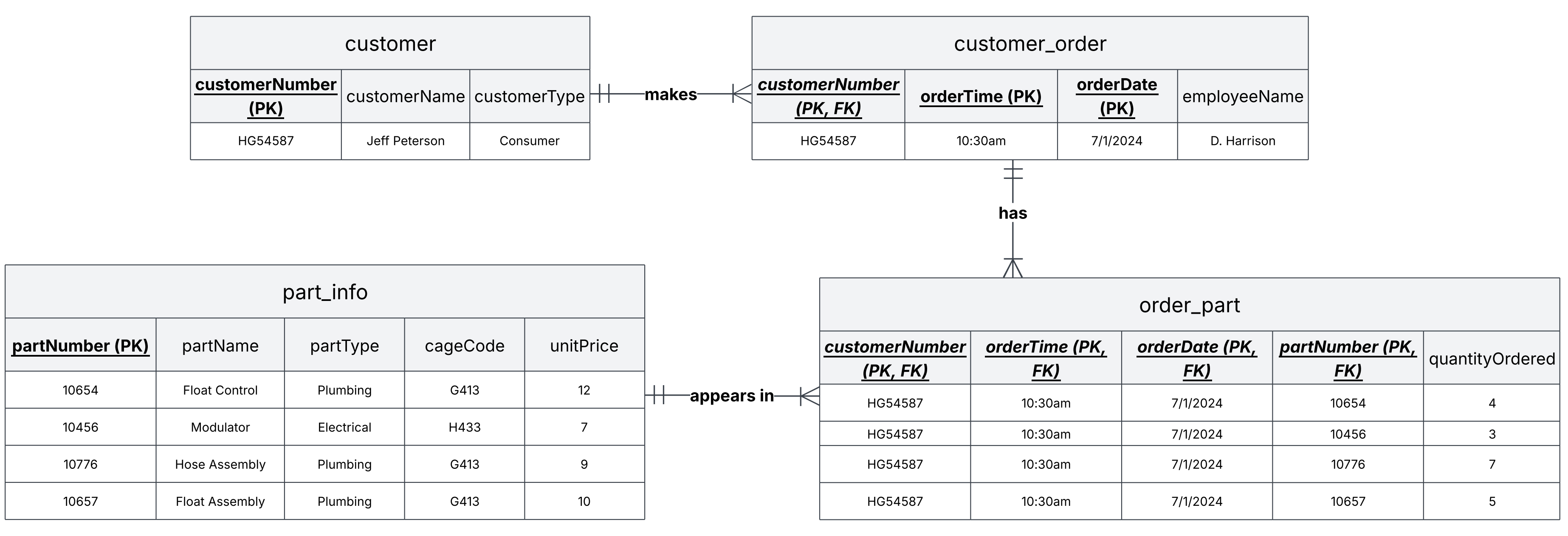
customerNumber (PK)	customerName	customerType
HG54587	Jeff Peterson	Consumer

customer_order

customerNumber (PK)	**orderTime (PK)**	**orderDate (PK)**	employeeName
HG54587	10:30am	7/1/2024	D. Harrison

I checked for transitive dependency, but there're none. So the 2NF relations are already in 3NF. Now I just need to identify the FK and create the final ERD.

Happy Supplies Parts Order Management ERD



customer

customerNumber (PK)	customerName	customerType
HG54587	Jeff Peterson	Consumer

customer_order

customerNumber (PK, FK)	**orderTime (PK)**	**orderDate (PK)**	employeeName
HG54587	10:30am	7/1/2024	D. Harrison

part_info

partNumber (PK)	partName	partType	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

order_part

customerNumber (PK, FK)	***orderTime (PK, FK)***	***orderDate (PK, FK)***	***partNumber (PK, FK)***	quantityOrdered
HG54587	10:30am	7/1/2024	10654	4
HG54587	10:30am	7/1/2024	10456	3
HG54587	10:30am	7/1/2024	10776	7
HG54587	10:30am	7/1/2024	10657	5

Assumptions: staffNo uniquely identifies a therapist and patNo uniquely identifies a patient. Since a therapist cannot have more than one appointment at the same date and time, so the unique appointment PK would be staffNo, appointmentDate and appointmentTime combined. Also since a therapist sees patients at only one branch on a given day, staffNo and appointmentDate would identify the branchNo.

The original table is already in 1NF since all attributes are atomic values and there are no repeating groups. The PK is the combination of staffNo, appointmentDate and appointmentTime.

therapist_track_original

staffNo (PK)	**appointmentDate (PK)**	**appointmentTime (PK)**	patNo	patName	therapistName	branchNo
S1011	9/12/2022	10:00	P100	Lily White	Fred Smith	M15
S1011	9/12/2022	12:00	P105	Jill Baker	Fred Smith	M15
S1024	9/12/2022	10:00	P108	Andy McKee	Heidi Pierce	Q10
S1024	9/14/2022	14:00	P108	Andy McKee	Heidi Pierce	Q10
S1032	9/14/2022	16:30	P105	Jill Baker	Richard Levin	M15
S1032	9/15/2022	18:00	P110	Jimmy Winter	Richard Levin	B13

However, there are partial attributes. therapistName is dependent on staffNo. branchNo is dependent on the combination of staffNo and appointmentDate, but not appointmentTime.
So here I removed the partial dependencies to create the 2NF.

therapist

staffNo (PK)	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

therapist_branch

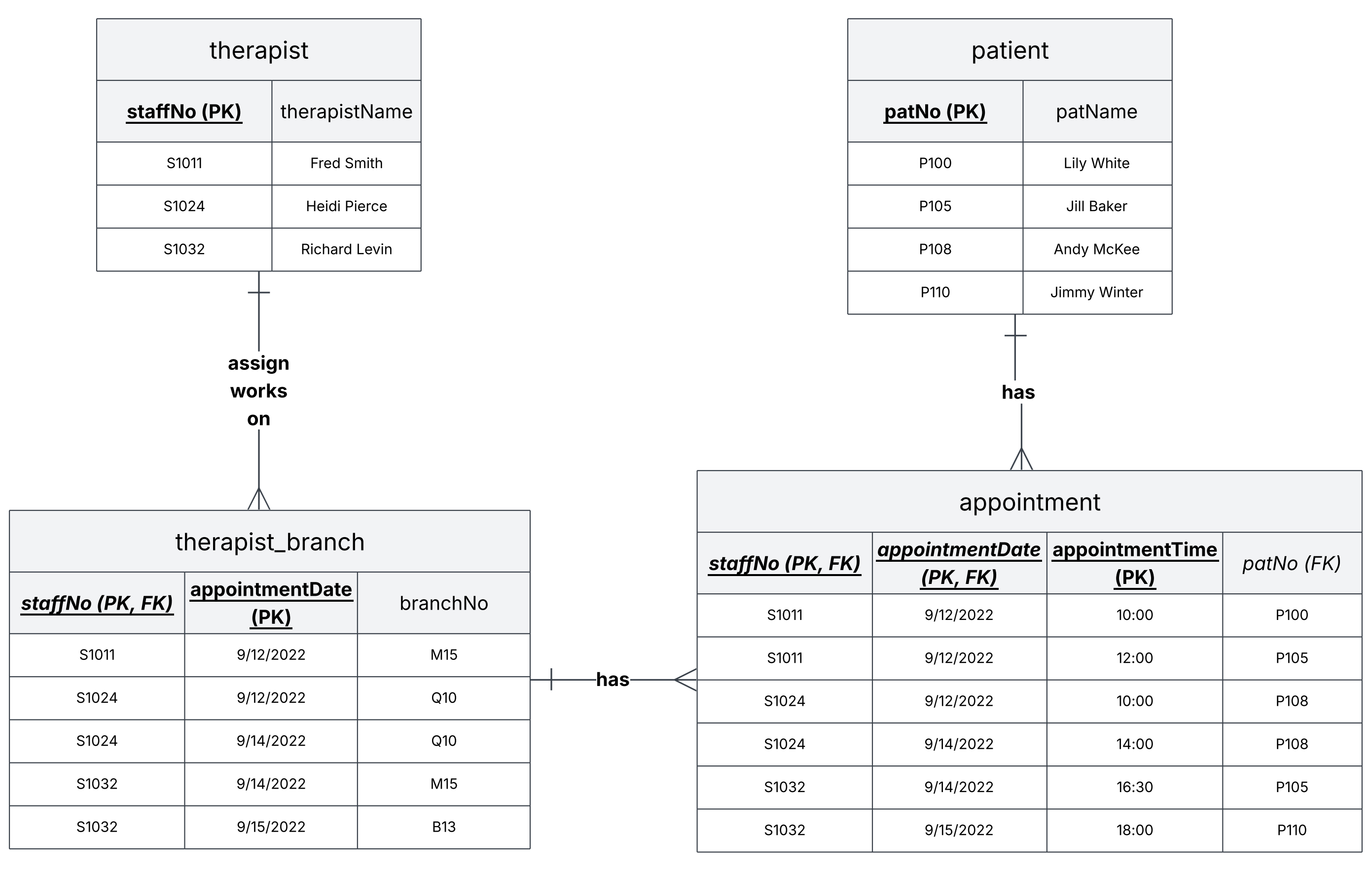
staffNo (PK)	**appointmentDate (PK)**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

appointment

staffNo (PK)	**appointmentDate (PK)**	**appointmentTime (PK)**	patNo	patName
S1011	9/12/2022	10:00	P100	Lily White
S1011	9/12/2022	12:00	P105	Jill Baker
S1024	9/12/2022	10:00	P108	Andy McKee
S1024	9/14/2022	14:00	P108	Andy McKee
S1032	9/14/2022	16:30	P105	Jill Baker
S1032	9/15/2022	18:00	P110	Jimmy Winter

There are still transitive dependency because patNo is dependent on the combined PK, but patName is dependent on patNo. I created a separate table to make the final ERD.

Panacea Mental Health Therapist Tracking ERD



therapist

staffNo (PK)	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

patient

patNo (PK)	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

therapist_branch

staffNo (PK, FK)	**appointmentDate (PK)**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

appointment

staffNo (PK, FK)	***appointmentDate (PK, FK)***	**appointmentTime (PK)**	*patNo (FK)*
S1011	9/12/2022	10:00	P100
S1011	9/12/2022	12:00	P105
S1024	9/12/2022	10:00	P108
S1024	9/14/2022	14:00	P108
S1032	9/14/2022	16:30	P105
S1032	9/15/2022	18:00	P110

Assumptions: eNo uniquely identifies an employee. contractNo uniquely identifies a contract. Each contract belongs to exactly one event, and the location is based on the event.

The original form is 1NF because there is no repeating groups and there is PK and atomic values. The PK is the combination of eNo and contractNo.

original_employee_contract_tracking

eNo (PK)	**contractNo (PK)**	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

However, There are partial dependencies because eName is dependent only on eNo, and event is based on contract, not employee. I separated the partial dependency into another table so that it's now 2NF.

employee

eNo (PK)	eName
1135	Smith J
1057	Hocine D
1068	White T

contract

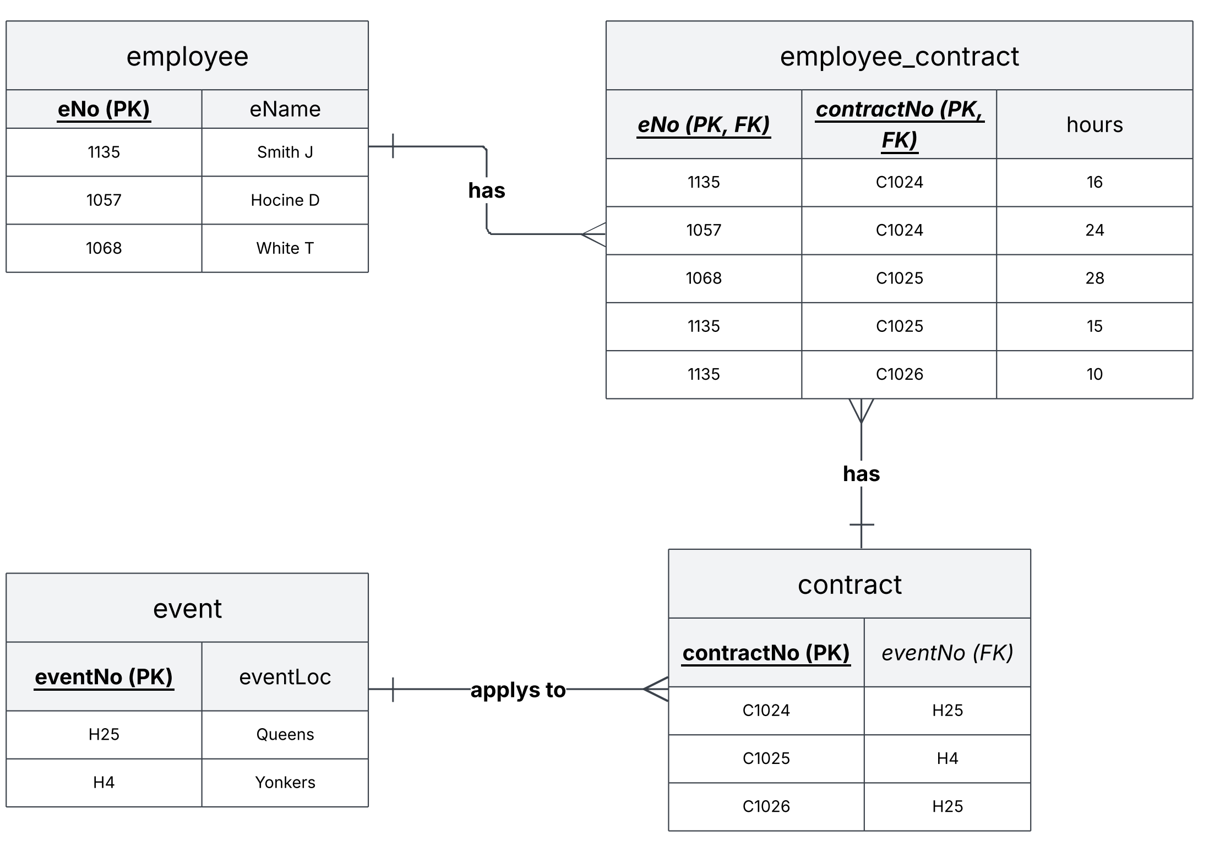
contractNo (PK)	eventNo	eventLoc
C1024	H25	Queens
C1025	H4	Yonkers
C1026	H25	Queens

employee_contract

eNo (PK)	**contractNo (PK)**	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

Here, I still need to remove transitive dependencies to create 3NF. eventNo is based on contractNo, but eventLoc is only dependent on eventNo.

NYC Maid Better temp agency Employee Hours Tracking ERD



employee

eNo (PK)	eName
1135	Smith J
1057	Hocine D
1068	White T

employee_contract

eNo (PK, FK)	***contractNo (PK, FK)***	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

event

eventNo (PK)	eventLoc
H25	Queens
H4	Yonkers

contract

contractNo (PK)	*eventNo (FK)*
C1024	H25
C1025	H4
C1026	H25